Locks

Locks are a form of secure bolt or clasp that require a specific key to open them. The key can be a physical object, a digital or numeric code, or a particular unique physical feature of a person. Among the most commonly used locks are cylindrical tumbler locks and combination locks.

Cylindrical tumbler locks

Commonly found in door fasteners and many padlocks, a tumbler lock consists of a barrel holding a cylinder—also known as a plug—that is able to turn. A series of chambers each contains a spring and pins of different lengths that prevent the cylinder from turning unless the correct key is inserted through an opening called the keyway.

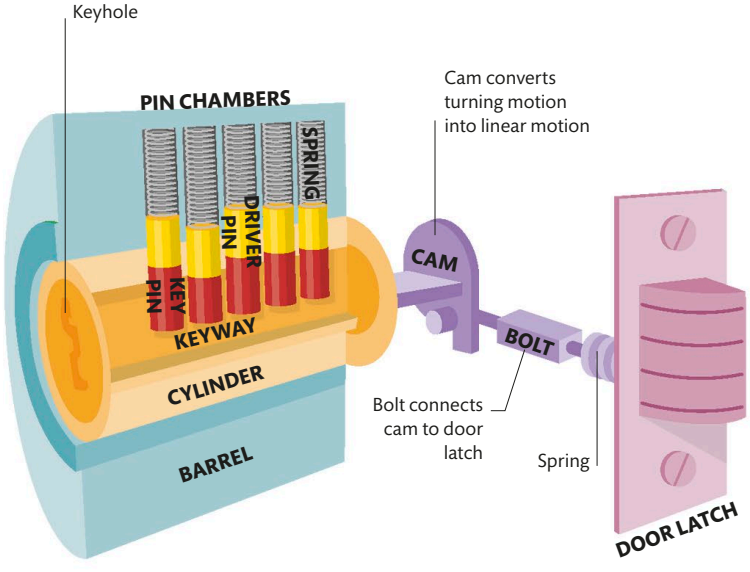
CAN YOU PICK A LOCK USING HAIRPINS?

In some simple tumbler locks, it is possible to use a combination of hairpins or wires to push up all the pins and turn the cylinder.

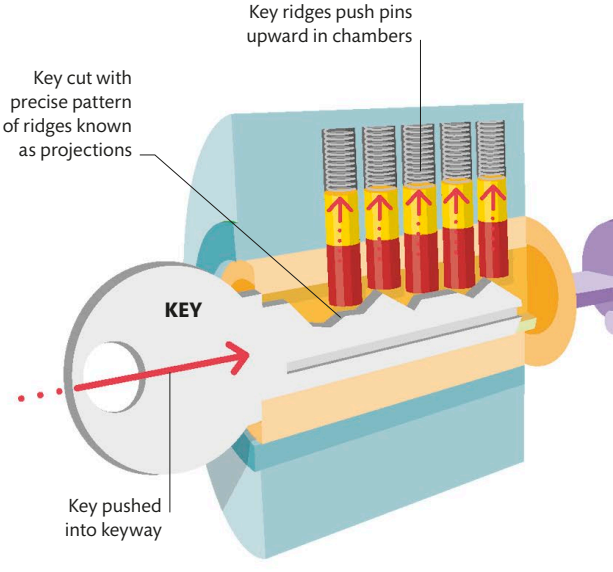


3 ft (90 cm)

THE **LENGTH** OF **THE KEYS** USED TO OPEN AND CLOSE THE **BOMBPROOF DOOR** TO THE BANK OF ENGLAND'S GOLD VAULT



1 **Lock closed**
In the locked position, the pins are pushed down their chambers by springs. This prevents the cylinder from turning, and the lock is closed.



2 **Key inserted into lock**
The key's ridges push the pins up precisely so that the tops of all the key pins align with the top edge of the cylinder.